

Ensuring the Timely and Orderly Interconnection of Large Loads) **RM26-4-000**

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With 1.7 GWhs of operating and in-construction battery energy storage and a development pipeline across the US of over 25 GWh of projects, esVolta is one of the largest and preeminent standalone storage developers, owners, and operators in the US. esVolta is a US-based company headquartered in Newport Beach, CA,¹ with operations throughout the US. We have invested over \$1B in storage projects to date and are committed to continuing to invest in regional markets where there is a resource adequacy and reliability need.

There is a place for “all of the above” new resources on the grid, but standalone energy storage is flexible, clean, cost-efficient, and ready to deploy quickly, making it the ideal technology to support rapid growth of data centers and meeting reliability and resource adequacy needs of the Bulk power system.

I. INTRODUCTION

On October 23, 2025, pursuant to authority under section 403 of the DOE Act, the Secretary of Energy directed the Commission to initiate rulemaking procedures and consider an ANOPR, which presented potential reforms to ensure the timely and orderly interconnection of large loads to the transmission system.² On October 27, 2025, the Commission issued a Notice Inviting Comments in response to the proposed ANOPR.

esVolta supports FERC exploring innovative approaches to enhancing reliability and managing affordability — all while working to promulgate new rules and regulations to allow for the transformation of our nation’s aging electricity grid to accommodate newer technologies, novel load configurations, and support national security and national interests. With an evolving resource

¹ esVolta is owned by Generate Capital PBC, an experienced and well-capitalized energy and infrastructure investor.

² *Ensuring the Timely and Orderly Interconnection of Large Loads*, RM26-4 (2025) (“ANOPR”).

mix and system usage, we need policies that evolve with the realities of the market to facilitate new system and market designs for 2025 and beyond.

We applaud FERC for being proactive on a number of policy fronts. For example, FERC issued Order 841 *to adopt reforms to remove barriers to the participation of electric storage resources and address “barriers to the participation of new technologies, such as many types of electric storage resources,³ in the RTO/ISO markets can emerge when the rules governing participation in those markets are designed for traditional resources and in effect limit the services that emerging technologies can provide.”⁴* Order 2023 enabled battery energy storage to fully participate in organized markets to provide reliability, resource adequacy, and affordability solutions. Order 841 envisioned this moment when the Commission directed the RTOs/ISOs to “recognize the physical and operational characteristics of electric storage resources, facilitate their participation in the RTO/ISO markets.... ensure that a resource using the participation model is eligible to provide all capacity, energy, and ancillary services that the resource is technically capable of providing.”⁵

In doing so, FERC has positioned BESS to serve large loads and support the Department of Energy’s intent behind the September ANOPR, considering that storage is flexible, dispatchable, cost-effective and curtailable. Whether FERC adopts the DOE ANOPR wholesale or modifies the language and principles outlined, it is appropriate that FERC continue the work it has already begun in select regions around these issues and facilitate a standardized national policy, process, and approach that builds off the great work done to expedite the interconnection of large loads and supporting generation and supply. The Commission should consider the below recommendations:

³ 162 FERC ¶ 61,127 18 CFR Part 35 [Docket Nos. RM16-23-000; AD16-20-000; Order No. 841] Electric Storage Participation in Markets Operated by Regional Transmission Organizations and Independent System Operators (Issued February 15, 2018)

⁴ Id at 6

⁵ Id at 8

(1) Energy storage is a solution to the projected electricity demand increase due to large load additions (2) Standalone battery energy storage offers the flexibility, efficiency, dispatchability and curtailability needed to facilitate the integration of large loads; (3) Standalone battery energy storage can ensure “speed to power” with the ability to be commercially operational within 12-18 months from interconnection; (4) Regional RTOs/ISOs have developed approaches that the Commission should consider in developing a national approach, such as allowing for front of the meter energy storage configurations to serve large loads and be eligible for expedited interconnection processing; and (5) Standalone battery energy storage can help minimize projected cost increases associated with integrating large loads.

II. COMMENTS

A. Electricity Demand Is Projected To Increase Exponentially Due To Large Load Additions And Standalone Storage Is A Solution

While U.S. electricity consumption was essentially flat for the last two decades, the electricity demand picture has changed dramatically in just the last two years. Grid operators have been sounding the alarm of the incongruent supply and demand outlook leading to potential supply deficits, and while there are many contributing factors, e.g., rapid deactivations of thermal generation, slow pace of entry of new supply resources, increasing electrification, and onshoring of manufacturing, and population growth, it is the unprecedented projected electricity demand for the integration and expansion of large load additions, namely data centers to support national security and American Leadership in Artificial intelligence, driving this unprecedented electricity demand growth.

Artificial intelligence is a national priority integral to national security and American economic leadership. The Administration recently released its America AI Action Plan, which underscores that winning the global “AI race” will require not only rapid innovation in models and

applications, but also an unprecedented buildout of domestic AI infrastructure, chips, data centers, and “the energy to power it all.” Upstream of data center buildout is the energy needed to power it all. The Action Plan explicitly recognizes that AI is the first digital service in modern life that demands “vastly greater energy generation than we have today” and calls for streamlined permitting for data centers and associated energy infrastructure, as well as a comprehensive strategy to “develop a grid to match the pace of AI innovation,”⁶ including stabilizing today’s grid, optimizing existing assets, and prioritizing the interconnection of reliable, dispatchable power resources.

Given this framework, esVolta agrees that there is a compelling need for the Commission to adopt reforms as outlined in the Department’s Letter initiating this ANOPR. Absent clear, standardized procedures for large load and hybrid interconnections, transmission providers will continue to rely on ad hoc approaches that risk delaying critical projects, exacerbating reliability concerns, and shifting costs to non-participating customers in ways that are inconsistent with open access and just and reasonable rate requirements. With the initiation of a comprehensive and transparent rulemaking process, the Commission can continue the work it has started and provide the regulatory certainty necessary to support national AI and manufacturing priorities, while protecting grid reliability and existing customers. If an ANOPR is issued, it should seek to include a proposal that articulates core principles, such as studying large loads and hybrid facilities together with generation where practicable, recognizing curtailable and dispatchable configurations (including those supported by BESS) for expedited treatment, and aligning cost responsibility with the specific injection and withdrawal rights requested.

⁶ Executive Order 14179, “Removing Barriers to American Leadership in Artificial Intelligence,” and “Winning the Race, America’s AI Action Plan (July 2025)

With peak electricity demand expected to grow 14% by 2030 and 54% through 2050⁷ this enormous appetite for electricity creates an opportunity to accelerate the build out of energy infrastructure. BESS in particular can provide essential reliability and resource adequacy services when located in general proximity to these data center facilities and the utilities that serve them. There are enormous opportunities for standalone BESS to provide grid supporting services and fill a supply gap that is widening as more data centers seek to come online and fuel economic development and our nation's global dominance in artificial intelligence (AI), hyperscale computing, and other large loads technologies and esVolta stands ready to support the data center buildout as part of an "*all of the above*" energy strategy.

1. BESS Can Enhance Power Quality to Assist in Facilitating the Timely and Orderly Interconnection of Large Load Consistent with the Principles of Reform

As the Department correctly recognizes, continued U.S. leadership in AI and the digital economy will depend in significant part on the timely development and interconnection of new data centers and other large, power-intensive computing facilities. esVolta supports this growth and agrees that Commission-jurisdictional transmission providers should facilitate the orderly integration of such large loads in a manner consistent with open-access and reliability obligations. At the same time, this new class of high-density, power-electronics-rich demand introduces distinct power quality and system performance challenges that legacy interconnection and planning frameworks were not designed to address.

In esVolta's view, the Commission's large-load reforms present an important opportunity to recognize that BESS can directly mitigate these power quality concerns and thereby assist in facilitating the timely and orderly interconnection of large loads consistent with the principles of reform outlined in this proceeding.

⁷ ICF Insights/Energy, "Rising current: America's growing electricity demand", [U.S. Demand Growth Forecast | ICF](#)

Data centers are known to display large load spikes and immediate drops in the matter of seconds due to dramatic rises in server rack power density. Recent reporting based on Bloomberg analysis underscores that large AI data centers are already degrading power quality on portions of the U.S. grid.⁸ The analysis found a strong correlation between proximity to data centers and elevated “bad harmonics,” with the most severe power-quality distortions concentrated in areas of dense data center development. At the same time, ERCOT and national laboratories have documented that sub synchronous oscillations from data center training can cause catastrophic turbine damage to facilities in close proximity. Data Centers tend to exhibit steep, short-duration changes in demand and highly power-electronics-intensive load profiles that can cause voltage excursions, frequency deviations, flicker, and other power quality issues at the point of interconnection. If not mitigated, these impacts can degrade service to neighboring customers and generation, and in many cases, necessitate more extensive and costly network upgrades to ensure continued compliance with applicable reliability standards and Open Access Transmission Tariff (“OATT”) requirements.

BESS is particularly well-suited to address the power quality challenges associated with large, rapidly changing loads such as AI data centers and high-performance computing facilities. Standalone BESS also can furnish a suite of power quality and reliability services that directly advance the principles for reform identified in this proceeding, including maintaining reliability, maximizing the efficient use of existing infrastructure, and supporting just and reasonable cost allocation.

Because BESS interfaces with the grid through advanced inverters and digital controls, it can function as a fast, controllable buffer between the large load and the bulk power system. When properly configured, BESS can be dispatched to smooth the net load seen by the transmission

⁸ Zachary Skidmore, *AI Data Centers Causing “Distortions” in US Power Grid – Bloomberg*, DATA CENTER DYNAMICS (Jan. 6, 2025), <https://www.datacenterdynamics.com/en/news/ai-data-centers-causing-distortions-in-us-power-grid-bloomberg/>

system, mitigating sudden spikes and drops in demand and thereby transforming a volatile large load into a more “grid-friendly” net profile. In practical terms, this enables transmission providers and RTOs/ISOs to study and plan around a more stable net injection/withdrawal pattern at the point of interconnection, which in turn can reduce the scale, cost, and lead time of network upgrades required to accommodate the same nominal level of load.

2. Battery Energy Storage is Electron and Fuel Agnostic Presenting Power Source Flexibility for Data Centers

BESS provides data center and grid operators alike with the needed flexibility as it is electron and fuel agnostic, curtailable, and dispatchable pairing well with all other energy generation resources, including on the power grid and/or hosted at a data center’s site. BESS is electron-agnostic in that it charges or stores generation produced by all resources—natural gas, coal, and nuclear generating facilities, as well as solar and wind. BESS discharges the stored energy back onto the grid when energy demand is high and the transmission system is experiencing constraints. Battery energy storage is a critical resource relied on to maintain resource adequacy and reliability.

BESS also complements all resource types, for example, BESS can enhance the efficiency and flexibility of natural gas plants by responding to dispatch orders to provide ancillary services, allowing thermal resources the ability to offer output into consistent need, high-demand energy markets, minimizing continuous ramping of the thermal facilities. The continuous ramping up to offer short spurts of energy provides needless wear and tear on the equipment, causing more occurrences of maintenance outages and ultimately decreasing the overall life of the thermal asset.

For example, in Texas, ERCOT’s CEO Pablo Vegas mused about this very issue:

“In most of the summer, wind has ramped up as solar output diminished at nightfall, which "has aligned well with the demand profile," Vegas said. Also, the state's thermal dispatchable fleet has delivered "a very strong and steady performance," he said ...but "Coming off a long stretch of very high demand and high utilization, it's not surprising to

see some mechanical breakages happening on some of the dispatchable generation fleet," Vegas said.⁹

3. Battery Energy Storage Can Ensure “Speed to Power” With the Ability to Be Commercially Operational Within 12-18 Months from Interconnection

RTO/ISOs have made significant improvements in the timelines for processing and studying interconnection request, but the immediate need for power to fuel data centers cannot wait through the normal processing timeframes. In supporting the President’s “energy dominance” and “AI” agenda, esVolta supports expedited interconnection processes for all of the above resources that have agreements with large loads.

Battery energy storage is a proven solution to meeting near-term and longer-term resource adequacy and reliability needs, with a modular design that can be right sized based on need, and one of the shortest construction time frames of any resource (12-18 months). As the nation faces imminent resource adequacy and reliability shortfalls, flexible resources that can be deployed in the near term, such as battery energy storage, are essential and must be properly valued for their speed to market. In contrast, it was mentioned at the technical conference that, on average, gas plants and other resources will take approximately five+ years.

With more than 12 GWs of energy storage resources added to the grid in 2024, storage has become the fastest-growing and most rapidly deployed energy infrastructure, it flexible in its configuration, and has proven its ability to deploy quickly.¹⁰ Given its fast deployment, data centers would benefit from the ability to share interconnection rights with existing storage interconnection rights.

⁹ [Markham Watson](#), Thermal generation outages contribute to recent Texas power market tightness

¹⁰ American Clean Power, **Energy Storage in MISO Report, ACP 2024**, <https://cleanpower.org/wp-content/uploads/2025/04/MISO-Storage-Reforms-4.6.25.pdf>

B. Load and Hybrid Facilities Should Be Studied Together With Generating Facilities

The ANOPR properly recognizes that there would be efficiencies gained in studying load and hybrid facilities together with identified generating facilities.¹¹ Studying large loads together with BESS aligns closely with the Department's emphasis on the timely and orderly interconnection of large loads. Where a large load is electrically proximate to BESS that is designed to provide the services described above, the combined configuration is better able to satisfy applicable reliability and power quality criteria without imposing excessive incremental burdens on the transmission system.

From the perspective of transmission providers and RTOs/ISOs, such paired configurations are more straightforward to study and integrate because the BESS can be modeled as a controllable injection and withdrawal resource at the same point of interconnection that actively mitigates the grid impacts of the associated large load. BESS enables the large loads to be curtailable or flexible and should jointly be eligible for an expedited pathway to interconnection.¹²

C. esVolta Encourages the Commission to Consider Model Approaches Regions Are Taking that Allow for Front-of-the-Meter BESS Configurations to Serve Large Loads and be Eligible for Expedited Interconnection Processing

Standalone BESS projects deployed in front-of-the-meter configurations provide a multitude of benefits to our nation's Bulk Power System, i.e., balancing supply and demand, improving grid stability, serving as a backup power source, providing black start capabilities, obviating the need for costly transmission and distribution system upgrades, enabling energy

¹¹ ANOPR at P.20.

¹² ANOPR at P24.

dominance and independence, **and now supporting immediate power needs of data centers and other large loads.** However, the ANOPR uses the term “hybrid facility” as “[large loads] seeking to share a point of interconnection with new or existing generation facilities,”¹³ which could be viewed as prohibitive for BESS. We are sure that was not the intention as many regions are developing similar processes that are inclusive of battery energy storage and its many configurations.

To that end, the Commission should clarify that consistent with the Commission’s Large Generator Interconnection Procedures, it intends for energy storage to be accommodated and included in its definition of “hybrid facilities” within the context of any proceeding aimed at ensuring the timely and orderly interconnection of large loads. Additionally, as stated below, the Commission should also expand its view of the load-supply configurations and expressly state that front-of-the-meter BESS configurations are also eligible configurations to serve large loads as many data centers are actively pursuing such arrangements.

With that said, the Commission should consider the great work and novel approaches that many regions across the US are developing and implementing to address the fast-growing demand for interconnecting data centers and other large loads, for example, SPP’s High Impact Large Load Generation Agreement (HILLGA) and PJM’s Large Load Additions Critical Issues Fast Path (CIFP) Initiative. These approaches are being developed in response to the onslaught of requests from large loads, namely data centers, to interconnect quickly, however, the current state of the generation interconnection queues in these regions cannot meet the quick time frames needed for these industries. Thus, the regions have moved quickly to develop expedited processes that allow new large loads to bring their own supply/generation, so to speak. The approaches being finalized recognize the myriad of load and supply configurations that will best serve these data centers and

¹³ ANOPR at P12.

their continuous need for power. While a large focus had been on co-locating supply and generation on site with these data centers, the reality is that supply resources “in close proximity” or “near by” provide that same value to the data centers, while importantly also being an available critical resource to the larger power grid and other grid customers, and being available to support the grid during emergency conditions.

1. Standalone BESS Located Within the Same Load Deliverability Area or Local Resource Zone is Well-suited to Serve Large Loads and Receive Expedited Interconnection Study Treatment Under Any New Commission Policy

esVolta encourages the Commission to consider a similar approach in any rulemaking it promulgates to address the integration of large loads and ensure that standalone BESS either located “in close proximity”, “near by” the data center, or either electrically or co-located onsite with the data center or the large load is deemed an eligible supply resource to receive expedited interconnection study treatment along with the data center load. esVolta would go even further to support approaches in PJM that rightfully identify supply resources in the *same “load deliverability area or LDA”* as an eligible resource for the expedited interconnection process and similarly, in MISO, standalone storage and other resources that have agreements with large loads to provide needed resources and are located in the same *“local resource zone or LRZ”* should be deemed eligible for an expedited interconnection process completed within 90 days. esVolta also supports allowing data centers to share interconnection rights and integrate with energy storage interconnection requests to facilitate sharing of grid capacity.

D. Standalone BESS Can Help Minimize Projected Cost Increases Associated with Integrating Large Loads

Affordability is a paramount consideration and the integration of large loads has the potential to increase costs for non-large load customer classes and ratepayers. The value proposition for deploying BESS to Support Data Centers includes the cost savings BESS provides

when deployed to provide peak capacity to support the new large load's peak demand needs. In peak demand situations, when BESS is deployed along with other dispatchable resources, for example, thermal generation facilities, BESS can provide the more instantaneous system needs through ancillary services, e.g., ramp up, ramp down, operating reserves, etc., allowing thermal resources to avoid the wear and tear on their assets from continuously ramping up and down to serve short duration needs and instead provide energy, capacity and other continuous, real time resource needs.

In Texas, the ability of standalone BESS to provide the flexibility needed to respond to evolving market conditions and changing market price signals resulted in BESS units saving an estimated \$750 million in day-ahead market costs by fulfilling essential ancillary services and freeing up to 3GW of gas generation to meet critical energy needs and reduce prices.”¹⁴ over just three critical peak days during emergency resource-constrained periods.

In addition, in most regions, existing transmission systems often times operate below their rated capacity many hours of the day. It is reported that only a small fraction of daily hours capture most of the economic value of those lines as "50% of transmission congestion value com[es] from only 5% of hours."¹⁵ Further, allowing storage to serve large loads is a cost and time efficient way to maximize the utilization of the existing transmission network more so than increasing the capacity of existing transmission lines, which in the context of expeditiously connecting and serving large loads would be cost and time prohibitive.

¹⁴ As battery storage systems have been increasingly providing ancillary services in Texas, they have enabled an equivalent amount of natural gas generation to participate in the system, meaning that one GW of newly constructed batteries participating in the ancillary services markets have resulted in about a GW of gas-fired generation coming out of back-up services and moving into the energy markets during periods of system stress...BESS helped reduce costs while keeping the lights on. During the January 2024 winter freeze, BESS units saved an estimated \$750 million in day-ahead market costs by fulfilling essential ancillary services and freeing up to 3GW of gas generation to meet critical energy needs and reduce prices.” Aurora Energy Research, **Role of Battery Energy Storage Systems (BESS) in the ERCOT Market**, May 2024

¹⁵ US Department of Energy, National Transmission Needs Study, Page v (October 2025)

CONCLUSION

esVolta appreciates this opportunity to provide comments to this ANOPR. We look forward to working with the Commission on this effort to maintain our nation's energy dominance and leadership in this global AI race. To that end, esVolta reiterates our recommendations and takeaways for consideration as the Commission determines its pathway forward: (1) Energy storage is a solution to the projected electricity demand increase due to large load additions (2) Standalone battery energy storage offers the flexibility, efficiency, dispatchability and curtailability needed to facilitate the integration of large loads; (3) Standalone battery energy storage can ensure “speed to power” with the ability to be commercially operational within 12-18 months from interconnection; (4) Regional RTOs/ISOs have developed approaches that the Commission should consider in developing a national approach, such as allowing for front of the meter energy storage configurations to serve large loads and be eligible for expedited interconnection processing; and (5) Standalone battery energy storage can help minimize projected cost increases associated with integrating large loads.

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